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Silent Fallback as a Denial-of-Validation Attack

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Abstract

Classical denial-of-service attacks produce observable degradation: services fail, error pages render, alerting fires. An adversary disrupting an agentic system's **validation path** — the channels by which the agent confirms freshness, authority, or source-of-truth — may produce a different outcome. If the agent falls back to cached or inferred state under validation disruption, and continues presenting confident output despite the disruption, the attacker has achieved a ***denial-of-validation*** outcome with no corresponding denial-of-service signal. The agent serves; the operator trusts; the validation gap is silent.

This paper specifies denial-of-validation (DoV) as a distinct attack class, structurally compounding denial-of-service (STRIDE-D) with spoofing (STRIDE-S). The compound is operationally consequential: classic availability defenses detect service unavailability and either fail-over or alert; classic spoofing defenses verify source identity but not source-availability-state. Neither defense catches the case in which a source is unavailable yet the agent presents downstream output as if the source had been consulted. Section 3 catalogues three attack mechanisms (DNS disruption, API rate limiting, authentication revocation) each of which can be weaponized into a DoV exploit. Section 4 specifies the ***fail-visible doctrine*** as the structural defense: explicit degraded-mode signaling at the agent-output layer, mandatory human checkpoints on authority loss, and a refusal-of-graceful-degradation discipline that converts silent exploits into observable incidents. Section 5 places DoV within the threat-modeling canon and discusses framework integration.

The central claim is that ***availability is an integrity property in agentic systems***: a system that gracefully degrades to stale-state operation without surfacing the degradation is operating **correctly** by classical availability standards while operating **incorrectly** by integrity standards. Closing the gap requires governance-mandated fail-visible behavior, not better availability engineering.

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ABSTRACT

Classical denial-of-service attacks produce observable degradation: services fail, error pages render, alerting fires. An adversary disrupting an agentic system's *validation path* — the channels by which the agent confirms freshness, authority, or source-of-truth — may produce a different outcome. If the agent falls back to cached or inferred state under validation disruption, and continues presenting confident output despite the disruption, the attacker has achieved a **denial-of-validation** outcome with no corresponding denial-of-service signal. The agent serves; the operator trusts; the validation gap is silent.

This paper specifies denial-of-validation (DoV) as a distinct attack class, structurally compounding denial-of-service (STRIDE-D) with spoofing (STRIDE-S). The compound is operationally consequential: classic availability defenses detect service unavailability and either fail-over or alert; classic spoofing defenses verify source identity but not source-availability-state. Neither defense catches the case in which a source is unavailable yet the agent presents downstream output as if the source had been consulted. Section 3 catalogues three attack mechanisms (DNS disruption, API rate limiting, authentication revocation) each of which can be weaponized into a DoV exploit. Section 4 specifies the **fail-visible doctrine** as the structural defense: explicit degraded-mode signaling at the agent-output layer, mandatory human checkpoints on authority loss, and a refusal-of-graceful-degradation discipline that converts silent exploits into observable incidents. Section 5 places DoV within the threat-modeling canon and discusses framework integration.

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1. INTRODUCTION

1.1 THE INVISIBILITY PROBLEM

A denial-of-service attack on a classical web service produces user-visible symptoms: the service is unreachable, error pages render, monitoring alerts. The attack is loud, and the defender's response is calibrated to that loudness — rate-limit the attacker, scale the service, route around the disruption.

A denial-of-service attack on an agentic system's *validation infrastructure* may produce no user-visible symptoms at all. The agent continues to serve. The output continues to read as confident. The validation gap — the fact that the agent could not confirm freshness or authority before answering — is invisible to the operator unless the agent surfaces it explicitly. If the agent's default is to *gracefully degrade* to cached or inferred state without flagging the



degradation, the attack is silent, the operator is uninformed, and the adversarial outcome propagates unimpeded.

The invisibility is the attack. The adversary's goal is not to make the agent unavailable; it is to make the agent's validation unreliable while preserving the appearance of validated operation.

1.2 WHY EXISTING FRAMEWORKS MISS IT

DDoS defenses operate on service-availability metrics: latency, error rates, request throughput. They do not measure *validation-path completeness* for downstream agentic operations. An agent that loses validation but continues serving does not produce DDoS-detectable signals.

Spoofing defenses (TLS, signed responses, authentication chains) operate on source identity: is this response really from the source it claims? They do not detect the case in which the source was *not consulted at all* and the response is the agent's inference from cached state.

OWASP LLM Top 10 (LLM03 data/model poisoning, LLM07 insecure plugin design) covers related attack surfaces but does not formalize validation-disruption as a distinct class.¹ MITRE ATLAS covers training-data integrity and adversarial inputs but not the operational-time validation infrastructure.²

The gap is that DoV operates at a layer prior frameworks do not name: the *agent's confidence in its own state-of-knowledge*. When validation is disrupted, the agent's epistemic state degrades; when the agent continues presenting confident output despite the degradation, the gap propagates as confident misalignment in the cyb-01 sense.³

1.3 CONTRIBUTION

This paper contributes:

Defines denial-of-validation (DoV) as a distinct attack class compounding STRIDE-D and STRIDE-S.

Catalogues three weaponizable mechanisms (DNS disruption, API rate limiting, authentication revocation) for inducing DoV in agentic deployments.

Specifies the fail-visible doctrine as the structural defense, with operational requirements at the agent-output layer.

Distinguishes fail-visible from graceful-degradation as defensive postures with structurally different security properties.

¹ OWASP, *OWASP Top 10 for Large Language Model Applications* (2024), <https://owasp.org/www-project-top-10-for-large-language-model-applications/>.

² MITRE, *Adversarial Threat Landscape for Artificial-Intelligence Systems (ATLAS)* (2024), <https://atlas.mitre.org/>.

³ Justin H. Kuiper, CISSP, *Confident Misalignment as an Adversarial Attack Surface* (v0.1-seed, 2026), nonsequitur.tech/white-papers/confident-misalignment-attack-surface/.



2. BACKGROUND

2.1 SILENT DEGRADATION: THE PARENT FRAMING

The HGC³AE² parent paper identifies silent degradation as a structural failure mode of agentic systems: the system continues producing outputs while quality, accuracy, or alignment degrades, with no architectural signal that distinguishes a degraded-context output from a normal-context output.⁴ The parent paper's defense is the runtime verification surface (Skipjack mechanisms). This paper extends the framing to adversarial weaponization: silent degradation is not merely a passive risk; it is an exploitable surface.

2.2 VALIDATION INFRASTRUCTURE IN AGENTIC SYSTEMS

Agentic systems depend on *validation infrastructure* — the channels by which the agent confirms freshness, authority, or source-of-truth. Validation infrastructure includes:

DNS resolution for upstream API endpoints (the agent must resolve `api.example.com` to consult the upstream).

API authentication (the agent must present credentials accepted by the upstream).

Rate limits (the upstream may impose per-time-window quotas that the agent must respect).

Certificate validation (the agent verifies the upstream TLS chain).

Token freshness (OAuth or similar token-bearing schemes require periodic refresh).

Each of these is a potential disruption surface. Disrupting any of them prevents the agent from consulting the upstream; if the agent's design includes a *fallback path* that serves cached or inferred state without signaling the disruption, the disruption converts into a DoV exploit.

2.3 THE DOS + SPOOFING COMPOUND

DoV compounds two classical STRIDE categories:⁵

Denial of Service (D) — the validation channel is disrupted; the agent cannot consult upstream truth.

Spoofing (S) — the agent's output, presented with normal confidence, *impersonates* validated output. The operator's trust heuristic accepts it as if validation had occurred.

Neither category alone characterizes the attack. DoV requires both: the validation must be disrupted (D) AND the output must continue to present as validated (S). Single-category defenses miss the compound.

⁴ Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.

⁵ Adam Shostack, *Threat Modeling: Designing for Security* (Hoboken, NJ: Wiley, 2014).



3. ATTACK MECHANICS

3.1 MECHANISM 1: DNS DISRUPTION

An adversary positioned to disrupt DNS resolution for an agent's upstream endpoints can prevent the agent from reaching authoritative sources. DNS-disruption techniques include cache poisoning at intermediate resolvers,⁶ BGP hijacking targeting authoritative servers, and route-injection attacks at adjacent network layers.

The agent's response depends on its fallback design. If the agent treats DNS-resolution failures as transient and falls back to cached lookup results (or to inferred-from-context state), the DNS disruption converts into validation-path silence. The agent continues to serve; the validation was never performed; the operator has no signal that the upstream was unreachable.

Detection at the agent layer. Track per-upstream resolution success/failure rates. Aggregate failure rates spanning multiple operating cycles indicate validation-path disruption regardless of whether individual outputs were served.

3.2 MECHANISM 2: API RATE LIMITING

An adversary who can induce upstream rate-limiting (either by overwhelming the agent's own request volume or by inducing the upstream to throttle the agent) creates a similar disruption. The agent's requests fail with rate-limit responses; the agent's fallback path determines whether the disruption is silent.

This mechanism is operationally subtle because rate-limiting is *expected behavior* at the upstream — it is not pathological from the upstream's perspective. Detection at the agent layer requires distinguishing “rate-limited and waiting for the next window” from “rate-limited and serving from cache because the next window is unavailable.”

Detection at the agent layer. Track per-upstream rate-limit events and the agent's subsequent behavior. Outputs served during rate-limit windows must be flagged as *served-without-validation* unless the agent explicitly waited for window reset.

3.3 MECHANISM 3: AUTHENTICATION REVOCATION

The most consequential mechanism. An adversary who can revoke or invalidate the agent's authentication credentials to an upstream source — by inducing OAuth token revocation, by triggering credential-rotation events out of band, by causing certificate-validation chain failures — disconnects the agent from authoritative sources entirely.

If the agent's design includes session-token caching with permissive expiration handling, the agent may continue serving from session state long past actual authorization. The agent believes it is authorized; the upstream considers the credentials revoked; the operator sees confident outputs that have not been validated against current upstream authorization state.

Detection at the agent layer. Track authentication-token freshness. Any output served on a credential that has not been validated within an operator-pinned validity window must be flagged as *served-on-stale-auth*.

⁶ Steven M. Bellovin, “Using the Domain Name System for System Break-ins,” *Proceedings of the 5th USENIX Security Symposium*, 1995.



4. THE FAIL-VISIBLE DOCTRINE

4.1 DEFINITION

The fail-visible doctrine specifies that **any agentic output served while validation-path disruption is active must surface the disruption at the output layer**. The doctrine is operational, not aspirational: the agent's output must include a structured disruption signal that the operator's tooling consumes and surfaces.

The doctrine is the structural defense against DoV. Where the agent fails-visible, DoV exploits convert from silent attacks to observable incidents; where the agent fails-graceful (continuing to serve without surfacing disruption), DoV exploits remain silent.

4.2 THE VISIBILITY REQUIREMENT

Visibility is enforced at two layers:

Agent-side. The agent emits a structured disruption signal alongside any output served while validation paths are disrupted. The signal names: which validation path was disrupted, when the disruption began, what fallback state was substituted, and what confidence-degradation should be assumed.

Operator-side. Tooling consumes the disruption signal and surfaces it to the operator. A disruption that occurs but is not surfaced is operationally equivalent to no disruption signal at all; the entire chain must be observable.

The visibility requirement composes with the cyb-02 Context Integrity Baseline:⁷ CIB defines per-entry context-integrity properties; the fail-visible doctrine adds the *signal-on-degradation* behavior when CIB properties cannot be verified due to validation disruption.

4.3 WHY GRACEFUL DEGRADATION IS INSUFFICIENT

Graceful degradation is the engineering pattern of continuing to serve with reduced functionality under disruption. In availability-focused systems, this is correct: serving stale-but-plausible content beats failing entirely. In governed agentic systems, this is **structurally incorrect**: serving stale-but-plausible content without signaling the staleness *erases the validation gap that the operator needs to see*.

The shift in operating discipline is significant. Engineering culture often treats graceful degradation as a virtue; the fail-visible doctrine treats it as a security defect for agentic systems with consequential outputs. The reframing: *serving uncertain content is acceptable; serving uncertain content without signaling the uncertainty is not*.

4.4 HUMAN-GOVERNED DEGRADED-MODE CHECKPOINTS

Fail-visible signals require human-governed response protocols. When the agent emits a disruption signal, the architecture's response depends on the operator's pinned policy:

For *low-consequence* outputs, the operator may accept degraded-mode operation with continued autonomy.

⁷ Justin H. Kuiper, CISSP, *Context Integrity Failure as the Agentic Cache-Poisoning Analog* (v0.1-seed, 2026), nonsequitur.tech/white-papers/context-integrity-failure/.



For *high-consequence* outputs, the operator-pinned policy requires a checkpoint: degraded-mode triggers human-in-the-loop intervention; the agent halts decision authority until the operator either authorizes degraded-mode continuation or restores validation infrastructure.

The checkpoint cadence is governance-defined. Skipjack Principle 3 (explicit uncertainty signaling) and Principle 4 (continuous reconciliation)⁸ together operationalize fail-visible in the parent framework; this paper specifies the threat-model frame in which those principles defend.

5. FRAMEWORK PLACEMENT

5.1 NEW ATTACK CLASS

DoV is not adequately captured by existing taxonomies. STRIDE's compound D+S structure is correct but the compound is not enumerated as a named threat in standard STRIDE practice. MITRE ATLAS does not currently include the operational-time validation-disruption pattern; an ATLAS technique covering DoV would be a natural extension.⁹

OWASP LLM Top 10 covers adjacent surfaces (LLM03 poisoning, LLM07 plugin design) but not DoV directly. A future revision could include DoV as a top-10 risk; the practitioner consequence is significant for any production LLM deployment with upstream-validation dependencies.

5.2 RELATIONSHIP TO CYB-01 AND CYB-02

DoV composes with cyb-01 (confident-misalignment-attack-surface) and cyb-02 (context-integrity-failure) as follows:

cyb-01 says: confident misalignment is an adversarial capability that completes at the human trust layer.¹⁰

cyb-02 says: context is a security primitive with integrity, freshness, and authority properties.¹¹

cyb-03 (this paper) says: validation disruption produces silent context-staleness that the agent presents with normal confidence, completing as confident misalignment.

The three theses together form a compositional attack: validation disrupted (DoV) → context unverified (CIB fails) → confident-but-wrong output (CMAS exploit). The defense at any one layer mitigates partially; the defense across all three (fail-visible + CIB enforcement + uncertainty signaling) mitigates structurally.

8 Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.

9 MITRE, *Adversarial Threat Landscape for Artificial-Intelligence Systems (ATLAS)* (2024), <https://atlas.mitre.org/>.

10 Justin H. Kuiper, CISSP, *Confident Misalignment as an Adversarial Attack Surface* (v0.1-seed, 2026), nonsequitur.tech/white-papers/confident-misalignment-attack-surface/.

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6. LIMITATIONS AND CONCLUSION

6.1 LIMITATIONS

The fail-visible doctrine requires agent-side instrumentation that current LLM-deployment platforms rarely supply by default. Operationalizing fail-visible may require platform-level changes (not application-level patches), with associated coordination cost.

The doctrine's effectiveness depends on operator-side tooling that surfaces disruption signals reliably. Where the operator-side tooling is incomplete, fail-visible signals may go unread, producing a defense-on-paper that is not a defense-in-practice.

Validation infrastructure is heterogeneous. The three mechanisms in §3 are representative; novel disruption mechanisms targeting newer validation channels (vector databases, model APIs, retrieval pipelines) will require taxonomy extension.

6.2 CONCLUSION

Denial-of-validation is a distinct attack class with structural properties that compound STRIDE-D and STRIDE-S in ways neither category alone captures. The defense is fail-visible operation: any output served under validation disruption must surface the disruption at the output layer, with operator-side tooling that consumes and surfaces the signal. Graceful degradation is insufficient because it erases the validation gap; fail-visible degradation is structurally correct because it preserves the operator's ability to detect the disruption.

This is Thesis Three (cyb-03) of the HGC³AE² Cyber Pillar derivative family. Composes with cyb-01 (confident-misalignment-attack-surface) and cyb-02 (context-integrity-failure) into a three-layer attack/defense model where validation disruption, context-integrity loss, and confident-output presentation each contribute to the compound exploit.

REFERENCES

— end v0.1-seed draft —

Drafting state: v0.1 CYB-03 (chain v0.1 step 3). 6-section canon shape. PR count: 12 —
parent (HGC³AE²) + 11 inline references
(STRIDE/ATLAS/OWASP-LLM/DNS/OAuth/NIST/Chaos + 2 cross-thesis refs cyb-01, cyb-02).
Meets ≥12 floor.